

Product carbon footprint

[Product carbon footprint](#) (PCF) is the aggregate of manufacturing, transport and usage emissions plus those related to end-of-life management. PCFs provide customers with insights for more informed purchasing decisions. We employ two methodologies to capture end-to-end PCF impact across our product portfolio: Product Attribute to Impact Algorithm (PAIA) and the Dell PCF Calculator.⁸

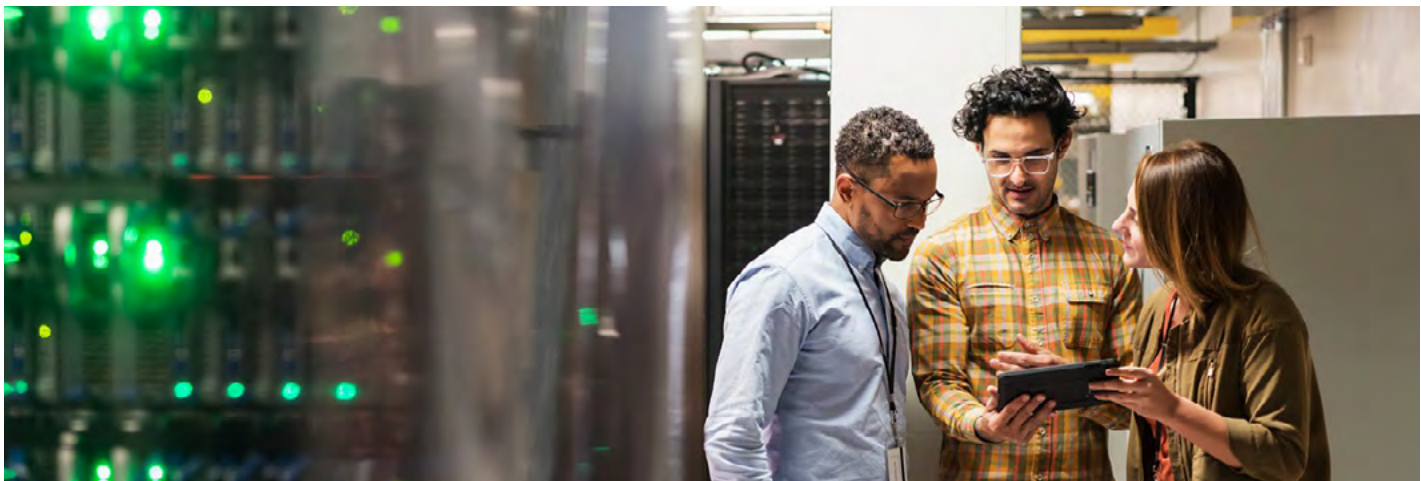
PAIA is commonly used throughout the consumer products industry. At Dell, we use PAIA for servers, storage, networking and legacy client products built prior to CY24. We will continue to use PAIA for infrastructure solutions group (ISG) servers, storage and networking client products.

The next step in our PCF calculation journey has been our development of the Dell PCF Calculator, which we have first applied to commercial client products. Several scope 3 emissions categories align with PCF data, including purchased goods and services, transportation and distribution and use of sold products.

Through our assessments and insights, we aim to reduce the carbon footprint of our products and solutions without compromising performance. We focus on reducing PCFs in four stages of the product lifecycle:

- **Manufacturing:** We actively reduce emissions from the start by using recycled, renewable and low-emissions materials.
- **Transportation:** By optimizing logistics and giving preference to regional ground shipping, we reduce delivery emissions.
- **Energy use:** We enhance efficiency in client solutions, servers and storage products to minimize energy consumption.
- **End-of-life management:** We responsibly recover, recycle and reuse retired assets to contribute to the circular economy.

More details on our work to reduce the impacts of our products throughout their lifecycles are in the [Circular Economy](#) section of this report.



Product energy efficiency

Eco label standards

We adhere to international [eco label](#) standards that guide our industry in areas of climate, circularity, transportation and energy efficiency. Eco labels help us build trust through third-party validations and make it easier for customers and partners to identify our most sustainable products.

Some of our products have earned ENERGY STAR® certifications and adhere to U.S. EPA energy-efficiency specifications. Over 500 of our computers, displays, enterprise servers, data center storage and networking equipment are ENERGY STAR-qualified. We also work in support of 80 Plus®, TCO and the China Environmental Label Program standards.



We participate in the [Global Electronics Council](#) EPEAT eco-labeling program where over 330 of our products are EPEAT-registered and more than half of those are EPEAT gold. EPEAT has a range of criteria for products to earn its label. EPEAT also launched a Climate+ designation specifically for products that meet stringent climate-related criteria. The EPEAT Climate+ designation assures customers their purchases have limited impact on climate change. In FY24, we were recognized as an EPEAT Climate+ Champion and an early adopter of the climate change criteria with over 200 registered Climate+ products. A list of product names and gold, silver, or bronze and Climate+ designations are on the [EPEAT registry](#).



Sustainable data centers

We work to improve our products’ energy efficiency through hardware, software and tailored efficiency solutions. Our technology and services allow us to partner with customers on their journey to build modern, sustainable data centers. We optimize our data center technology for efficient energy performance in two main areas:

- Our hardware is designed to deliver maximum performance per watt of energy used.
- Breakthrough software solutions like [OpenManage Enterprise](#) (OME) Power Manager and [CloudIQ](#) provide energy management insights.

The inefficient use of IT assets is a large source of excessive energy consumption in data centers. We have reshaped our understanding of thermals and cooling best practices over the years to help customers minimize energy waste. For example, by implementing variable speed fans and infrastructure purpose-built for higher temperatures, our customers can avoid over-cooling and waste less energy.

To improve data center efficiency and performance and deliver value to customers, we combine several elements:

- **Audit infrastructure:** We examine components for opportunities to minimize power demands. For example, we improve asset density to reduce floor space and the corresponding cooling required. Our OpenManage Enterprise (OME) power manager delivers telemetry to help lower the carbon footprint.

- **Right-sized footprint:** By installing only necessary assets, our customers can reduce wasteful over-provisioning. Right-sized IT assets either reduce the power draw for equipment, make the power draw efficient, or both. Dell’s PowerEdge power supply units (PSU) portfolio incorporates strategies to enable right-sizing of PSU capacity based on platform configuration and target workloads. Additionally, our [APEX](#) as-a-service (aaS) offerings provide customers with these right-sized IT resources and the ability to scale when needed.
- **Optimized HVAC functionality:** We engineer new ways to reduce heat generated by our powerful machines through cooling systems and thermal advancements. For example, our EPEAT Climate+ [PowerEdge servers](#) with direct liquid cooling systems minimize energy needed to cool the data center and reduce reliance on HVAC systems.
- **Intelligent platform management:** Our servers have built-in Basic Input or Output System (BIOS) and Integrated Dell Remote Access Controller (iDRAC) settings to manage off-peak power consumption. These intelligent platforms can also identify potential issues.
- **Optimized workload:** Our data center solutions help manage workloads on premises and in the cloud. Customers that transition to high-density and energy efficient flash storage experience substantial enterprise data storage reductions.

Learn more about how our [Energy Efficient Data Center](#) solutions are designed to increase efficiency while reducing energy costs and emissions.